Codveda Data Analytics Internship

Projects completed for the **Data Analytics internship at Codveda Technology**: two tasks at each of three levels, written in Python on publicly available datasets.

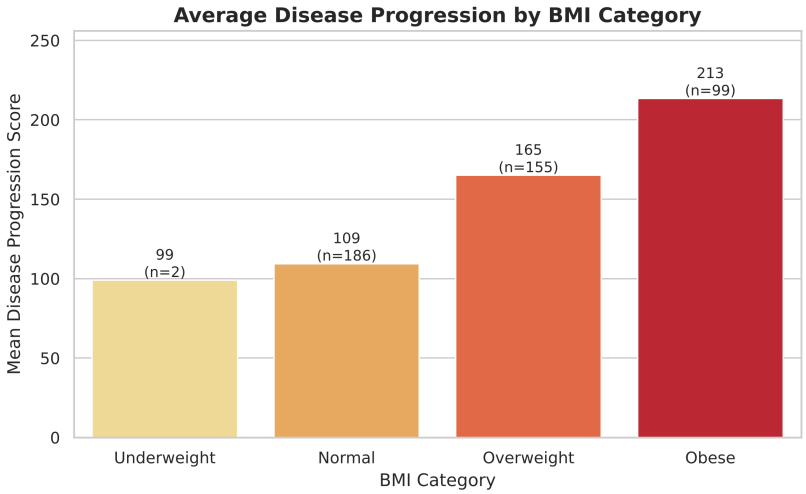
Projects

Level	Task	Dataset	Script
1 (Basic)	Exploratory Data Analysis	Diabetes progression, 442 patients	level1_eda_and_visualization.py
1 (Basic)	Basic Data Visualization	Same	Same file
2 (Intermediate)	Regression Analysis	Diabetes progression	level2_regression_and_clustering.py
2 (Intermediate)	Clustering (K-Means)	UCI Wine, 178 samples, 13 features	Same file
3 (Advanced)	Predictive Modeling (Classification)	Wisconsin Breast Cancer, 569 tumours	level3_classification.py
3 (Advanced)	NLP Sentiment Analysis	NLTK movie reviews, 2,000 reviews	level3_nlp_sentiment_analysis.py

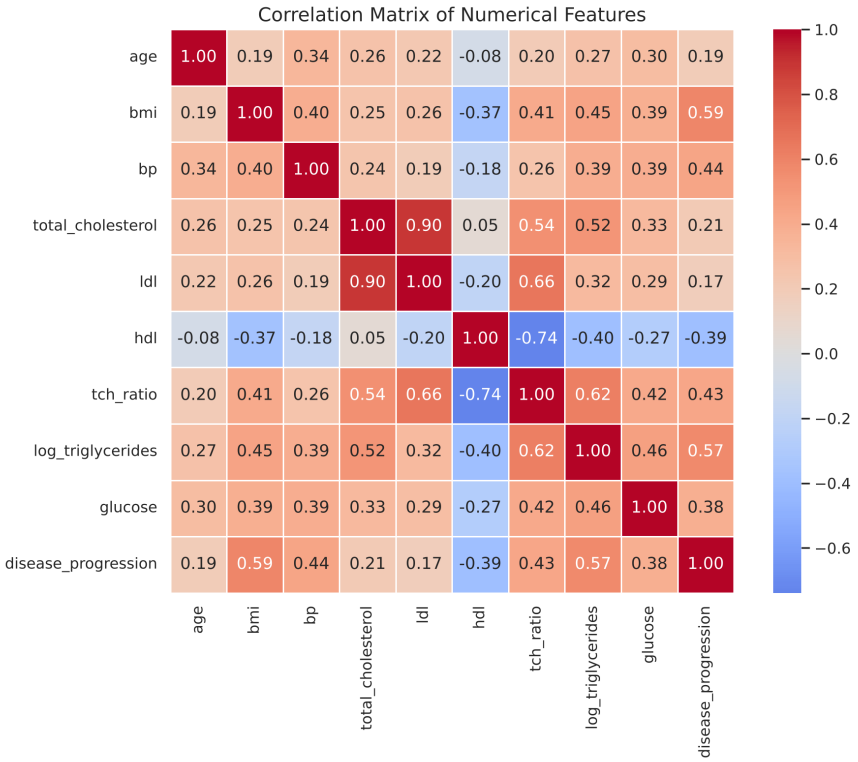
Results at a glance

Level 1: Exploring the data

- No missing values or duplicates in the dataset.
- BMI (r = 0.59) and log-triglycerides (r = 0.57) correlate most strongly with disease progression.
- Average progression rises with BMI category: Normal 109, Overweight 165, Obese 213.
- Total cholesterol and LDL are almost duplicates (r = 0.90), which matters for modelling.



Average disease progression by BMI category

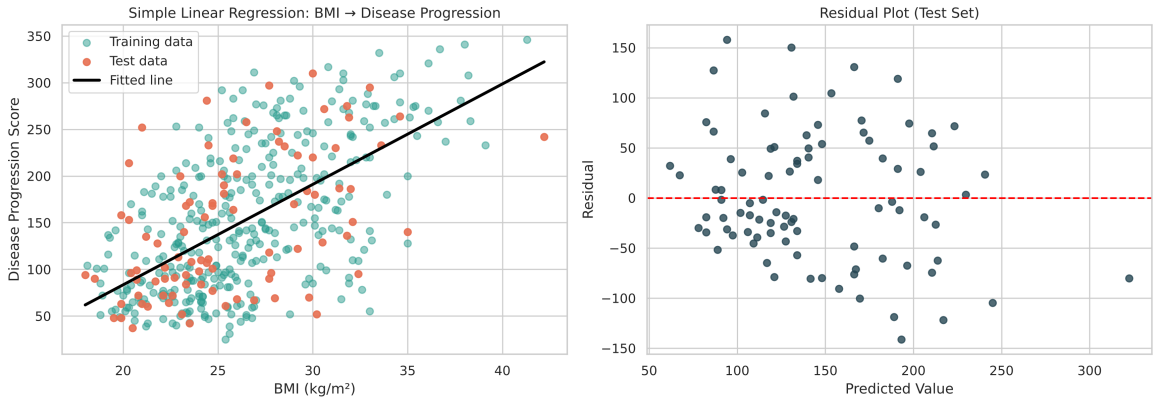


Correlation matrix of numerical features

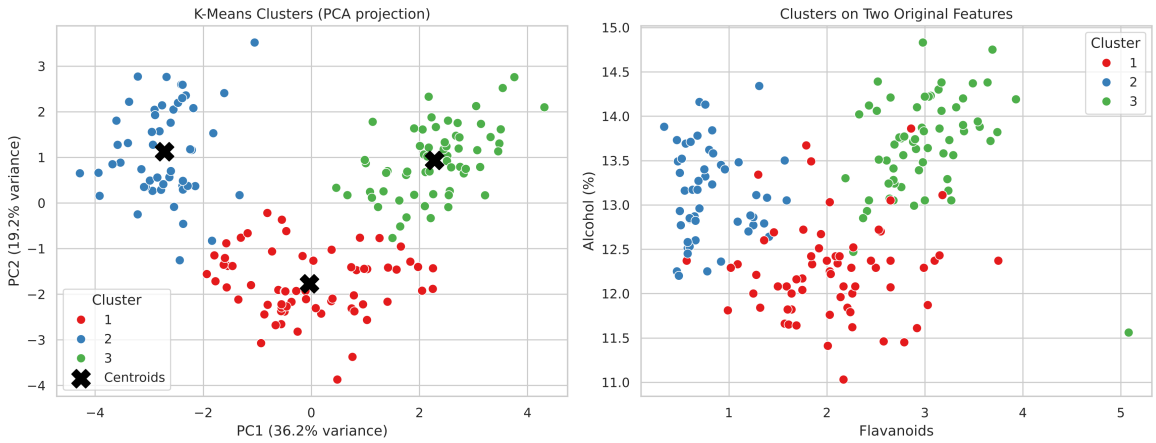
Level 2: Regression and clustering

- **Regression** (80/20 split): each extra BMI unit is associated with about 10.8 more points of progression. BMI alone gives a test R^2 of 0.23 (RMSE 63.7). Using all 10 features raises R^2 to 0.45.

- **K-Means** on standardised data: the elbow method and silhouette score both select k = 3. The clusters closely match the true wine cultivars (Adjusted Rand Index 0.90), although the labels were never used for fitting.



Simple linear regression: BMI to disease progression, with residual plot



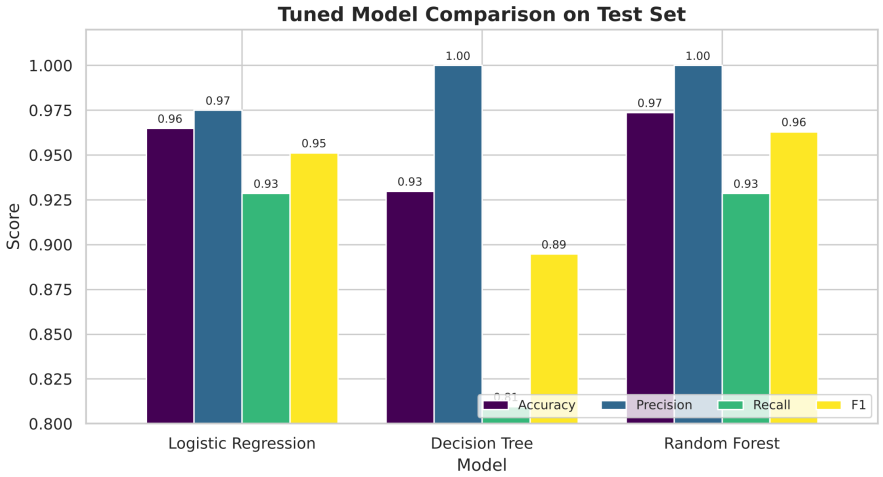
K-Means clusters (PCA projection and two original features)

Level 3: Classification

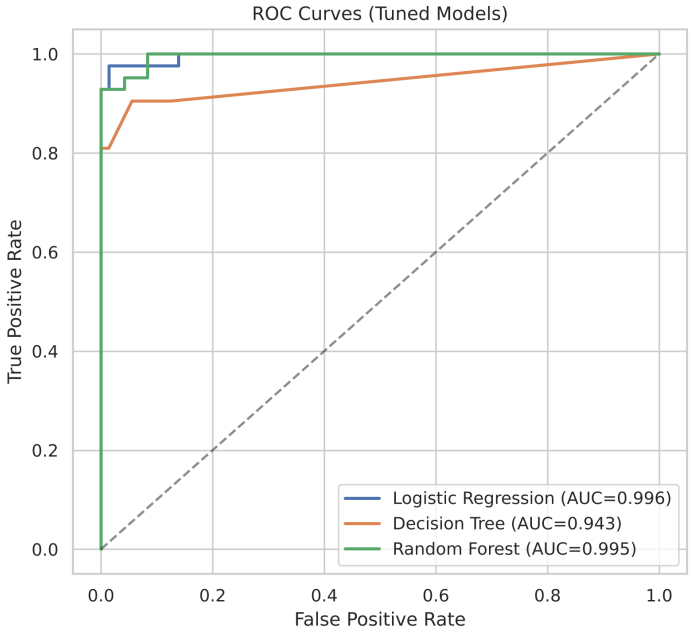
Positive class = malignant. Three models were trained on scaled features and tuned with 5-fold grid search (scored on F1). Test set: 114 tumours.

Model (tuned)	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.965	0.975	0.929	0.951	0.996
Decision Tree	0.930	1.000	0.810	0.895	0.943
Random Forest	**0.974**	**1.000**	0.929	**0.963**	0.995

Grid search did not improve test scores for Logistic Regression or Random Forest. With a small test set, these differences are within noise. In a medical setting recall matters most, because a missed malignant tumour is the costly error.



Tuned model comparison on the test set



ROC curves of the tuned models

How to run

```
pip install -r requirements.txt
python level1_basic/level1_eda_and_visualization.py
python level2_intermediate/level2_regression_and_clustering.py
python level3_advanced/level3_classification.py
python level3_advanced/level3_nlp_sentiment_analysis.py  # needs internet
```

Each script saves its charts to a **figures/** folder next to it, along with CSV tables and a results log.

Tools

Python, pandas, NumPy, scikit-learn, matplotlib, seaborn, NLTK, TextBlob, WordCloud.

Notes

- The datasets are classic teaching datasets bundled with scikit-learn or NLTK. In the diabetes data, sex is coded 1/2 with no documented meaning, so it appears as "Group 1/2".

- The NLP script needs internet access for its NLTK data; run it once in Google Colab to generate its outputs.

- Correlations describe patterns in the data and do not prove causes.

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